DATE:12-08-2026

EXPERIMENT 12

Tic-Tac-Toe Game in Python

Aim

To write a Python program to implement a **Tic-Tac-Toe game** for two players using a 3 × 3 game board.

Algorithm

1. Start the program.
2. Create a 3 × 3 board with empty spaces.
3. Display the board.
4. Ask Player 1 to enter a position and place **X**.
5. Ask Player 2 to enter a position and place **O**.
6. After every move, check whether a player has three matching symbols in a row, column, or diagonal.
7. If a player wins, display the winner and stop.
8. If all nine positions are filled without a winner, display **Draw**.
9. Otherwise, continue until the game ends.

Program

```
# Tic-Tac-Toe Game


board = [" " for _ in range(9)]


def display_board():
    print()
```

```
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])
    print()

def check_winner(player):
    winning_positions = [
        (0, 1, 2),
        (3, 4, 5),
        (6, 7, 8),
        (0, 3, 6),
        (1, 4, 7),
        (2, 5, 8),
        (0, 4, 8),
        (2, 4, 6)
    ]

    for a, b, c in winning_positions:
        if board[a] == board[b] == board[c] == player:
            return True
    return False
```

```
def play_game():
    player = "X"

    for turn in range(9):
        display_board()

        print("Player", player, "turn")
        position = int(input("Enter position (1-9): ")) - 1

        if position < 0 or position > 8 or board[position] != " ":
            print("Invalid position! Try again.")
            continue

        board[position] = player

        if check_winner(player):
            display_board()
            print("Player", player, "wins!")
            return

        if " " not in board:
            display_board()
            print("Game Draw!")
            return
```

```
        if player == "X":
            player = "O"
        else:
            player = "X"


play_game()
```

Input

The players enter a position from 1 to 9.

Player X turn

Enter position (1-9): 5

Player O turn

Enter position (1-9): 1

Output

```
  |   |
--+---+--
  |   |
--+---+--
  |   |

Player X turn
Enter position (1-9): 
```

Result

Thus, the Tic-Tac-Toe game was successfully implemented in Python, allowing two players to play alternately and detecting win, draw, and invalid moves.